MIL-E-18435C
16 April 1980
SUPERSEDING
MIL-E-18435B
19 June 1970

MILITARY SPECIFICATION

EXHAUST HEAD, PIPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers cast iron and steel exhaust heads for moisture and oil entrainment and discharge of exhaust steam into the atmosphere.

1.2 Classification. The exhaust heads shall be of the following types and sizes as specified (see 6.2):

Type I - Cast iron.
Type II - Fabricated steel plate.

Size	1	- 1-inch inlet connection.
Size	1-1/4	- 1-1/4 inch inlet connection.
Size	1-1/2	- 1-1/2 inch inlet connection.
Size	2	- 2-inch inlet connection.
Size	2-1/2	- 2-1/2 inch inlet connection.
Size	3	- 3-inch inlet connection.
Size	3-1/2	- 3-1/2 inch inlet connection.
Size	4	- 4-inch inlet connection.
Size	5	- 5-inch inlet connection.
Size	6	- 6-inch inlet connection.
Size	8	- 8-inch inlet connection.
Size	10	- 10-inch inlet connection.
Size	12	- 12-inch inlet connection.
Size	14	- 14-inch inlet connection.
Size	16	- 16-inch inlet connection.
Size	18	- 18-inch inlet connection.
Size	20	- 20-inch inlet connection.
Size	24	- 24-inch inlet connection.
Size	30	- 30-inch inlet connection.

1.2.1 Part number. The military specification part number for items described in this specification will be identified as shown in 6.6.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4730

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-775 - Sheet Steel, Zinc Coated.

MILITARY

MIL-V-3 - Valves, Fittings, and Flanges (Except for Systems Indicated Herein): Packaging of.

STANDARDS

FEDERAL

FED STD H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B16.1 - Cast Iron Pipe Flanges and Flanged Fittings, Class 125.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A278 - Standard Specification for Gray Iron Castings for Pressure-Containing Parts for Temperature up to 650°F (345°C).
A285 - Standard Specification for Pressure Vessel Plates, Carbon Steel, Low and Intermediate-Tensile Strength.

(Application for copies should be addressed to the American Society for Testing and Materials, 1016 Race Street, Philadelphia, PA 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

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3. REQUIREMENTS

3.1 Description. Exhaust heads shall consist of a cast or fabricated receiver with a flanged or threaded inlet connection, atmosphere exhaust port, and a threaded drain connection.

3.2 First article. When specified (see 6.2), the contractor shall furnish an exhaust head of the type and size as required for first article inspection and approval (see 4.2.1 and 6.4).

3.3 Standard commercial product. Each exhaust head of the same classification shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification, but which are a part of the manufacturer's standard commercial product, shall be included in the exhaust head being furnished. Standard commercial product is a product, which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specificallay specified.

3.4.1 Exhaust heads.

3.4.1.1 Type I heads. Type I exhaust heads shall be constructed of cast iron conforming to class 30 of ASTM A278.

3.4.1.2 Type II heads. Type II exhaust heads shall be constructed of steel plate conforming to grade C of ASTM A285.

3.4.1.2.1 Steel plate. When exhaust heads are constructed of steel plate less than 3/16 inch in thickness, the steel shall conform to class C of QQ-S-775.

3.4.2 Fasteners and washers. All fasteners and washers used in assembly shall be made of corrosion-resisting materials.

3.5 Design and performance. Exhaust heads shall satisfactorily remove entrained moisture and oil from exhaust steam, and shall satisfactorily suppress exhaust noise at least 50 percent under the working conditions specified (see 6.2). Removal of entrainment shall be affected by both deflection and expansion of steam. Means shall be provided within the exhaust head for collecting oil and moisture and preventing these liquids from

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rejoining and escaping with the steam. The volume of the receiver shall be adequate to allow rapid expansion of the steam for effective separation of entrainment. The area of the exhaust porting shall be greater than the area of the inlet connection in order to minimize back pressure.

3.5.1 Rating. Recommended exhaust head ratings shall be not less than indicated in Table I, and steam discharged to atmosphere at this rating shall not contain more than 0.5 percent moisture at the exhaust opening.

TABLE I. Exhaust head ratings.

Inlet connections Size in inches	Recommended minimum rating in pounds of steam per hour
1	70
1-1/4	120
1-1/2	160
2	260
2-1/2	370
3	570
3-1/2	770
4	990
5	1550
6	2240
8	3900
10	6100
12	8800
14	10700
16	14200
18	18100
20	22500
24	33000
30	51500

3.6 Size. Exhaust head size shall be designated by inlet connection size (see 6.2).

3.6.1 Inlet connection. Exhaust heads in size 4 inch and smaller shall have American National taper pipe threaded connections in accordance in FED STD H28. Exhaust heads in size 5 inch and larger shall be furnished with 125 pound ANSI standard flanges, faced and drilled to conform with the requirements of ANSI B16.1.

3.6.2 Drain connection. Drains shall be adequate to carry off all entrainment, and shall be not less than the sizes indicated in Table II.

TABLE II. Drain sizes.

Exhaust head size (inches)	Drain size (inches)
1, 1-1/4, 1-1/2	1/2
2, 2-1/2	3/4
3, 3-1/2, 4	1
5, 6	1-1/2
8, 10	2
12, 14	2-1/2
16, 18, 20	3
24, 30	4

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3.7 Identification marking. Equipment, assemblies, and parts shall be marked for identification in accordance with the manufacturer's standard practice.

3.8 Finish. The exhaust head shall be finished in accordance with the manufacturer's standard practice.

3.9 Workmanship.

3.9.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.9.1.1 Seams. All seams on fabricated exhaust heads shall be welded.

3.9.2 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

3.9.3 Bolted connections. All bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided. All nuts, bolts, and screws shall be tight.

3.9.4 Welding. Surfaces to be welded shall be free from foreign matter which would be injurious to the weld. Welding procedures shall be in accordance with a nationally recognized code. Welds shall be of sufficient size and shape to develop the full design strength of the parts connected by the welds. Welds shall transmit imposed stresses without deformation or failure when subjected to proof or service loadings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2.1).
(b) Quality conformance inspection (see 4.2.2).
(c) Packaging inspection (see 4.7).

4.2.1 First article inspection. First article inspection shall be performed on one complete exhaust head when a first article sample is required (see 3.2 and 6.4). This inspection shall include the examination of 4.5 and the tests of 4.6. The first article may be a standard production item from

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the contractor's current inventory, provided the exhaust head meets the requirements of this specification and is representative of the design, construction, and manufacturing technique applicable to the remaining exhaust heads to be furnished under the contract.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be performed on the sample exhaust heads selected in accordance with 4.4. This inspection shall include the examination of 4.5 and the tests of 4.6.

4.3 Inspection lot. All units of the same type and size offered to the Government at one time, shall be considered a lot for purpose of inspection. The sample unit shall be one complete exhaust head.

4.4 Sampling. A random sample of exhaust heads shall be selected from each lot in accordance with MIL-STD-105.

4.4.1 Sampling for examination. Examination of the exhaust heads shall be based on inspection level II, and an Acceptable Quality Level (AQL), of 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.2 Sampling for tests. Tests of the exhaust heads shall be based on inspection level S-3 and an AQL of 4.0 percent defective.

4.5 Examination. The first article, when furnished, and each sample selected in accordance with 4.4.2 shall be examined to verify compliance with this specification. Examination shall be conducted as specified in Table III.

TABLE III. Classification of defects.

Categories	Defect	Requirement paragraph
Major:		
101	Type of exhaust head not as specified.	1.2
102	Dimensions not as specified.	3.6
103	Cracks, blow holes, porosity, hard spots, or shrinkage in castings.	3.9.2
104	Threaded connections or flanges not as specified.	3.6.1
105	Welded surfaces not free of rust, scale, grease, or other foreign matter.	3.9.4
106	Kinks and sharp bends.	3.6.1
107	Seams not as specified.	3.9.1
108	Bolted connections not as specified.	3.9.3

4.6 Tests. The first article, when furnished, and each exhaust head selected in 4.4.2 shall be subjected to the tests specified in 4.6.1 and 4.6.2 to determine if the requirements of 3.5 and 3.5.1 have been met.

4.6.1 Noise suppression. The noise suppresion characteristics of exhaust heads shall be determined on a steam system at the applicable exhaust head rating shown in Table I. As steam escapes from a pipe at the applicable exhaust head rating (pounds per hour), the noise level of the escaping steam shall be measured by a suitable wide-band sound level instrument and recorded. The exhaust head unit shall then be placed on the top of the pipe

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from which the steam escapes. The steam shall then be allowed to flow through the exhaust head at the same rate as previously and the noise level shall be determined. The percent of noise suppression shall be determined as follows:

$$\text{Percent noise suppression} = \frac{N_1 - N_2}{N_1} \times 100$$

Where N_1 = Noise level meter reading of steam pipe without exhaust head.
N_2 = Noise level meter reading of steam pipe with exhaust head attached.

4.6.2 Moisture entrapment test. The exhaust head shall be subjected to the applicable steam rating specified in Table I. The moisture entrapped or condensed in the exhaust head shall be collected in a suitable container for a period of 10 minutes and weighed. The pounds of condensate collected per hour shall be determined as follows:

$$\text{Pounds of condensate per hour} = \text{pounds of condensate collected per 10 minutes} \times 6$$

If the amount of condensate (in pounds per hour) collected does not equal or exceed 99.5 percent of the value shown in Table I, the lot represented by the sample shall be rejected.

4.7 Packaging inspection. The preservation-packaging, packing, and marking of the exhaust heads shall be inspected to verify conformance to the requirements of Section 5.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. The exhaust heads shall be preserved, packaged, packed, and marked in accordance with MIL-V-3. The level of preservation and packaging and level of packing shall be as specified (see 6.2).

6. NOTES

6.1 Intended use. Exhaust heads covered by this specification are intended to remove water, oil, and other impurities from exhaust steam discharging to the atmosphere.

6.1.1 Selection of exhaust heads. When the exhaust steam rate is relatively constant, exhaust heads should be selected approximately in accordance with the ratings given in Table I in order to secure the most effective performance. When the exhaust steam rate is intermittent, it may be desirable to sacrifice optimum performance in favor of a smaller exhaust line. Carryover of entrainment should not be excessive and noise should not be objectionable for intermittent exhaust rates approximately twice the values shown in Table I. Discharge of unusually high presure steam, necessity for unusually quiet operation, or other difficult operating conditions require individual consideration, however, and should be fully stated in the ordering data.

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6.1.2 Sizes. Type I exhaust heads are available in all standard pipe sizes from 1 inch to 30 inches. Type II exhaust heads are fabricated in all standard pipe sizes from 1 inch to 30 inches. Sizes over 30 inches may be obtained on a custom design basis.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2 and 6.1.1).
(c) Size required (see 1.2, 2.6, and 6.1.2).
(d) Maximum exhaust steam rate in pounds per hour (see Table I).
(e) If a first article is not required (see 3.2 and 4.2.1).
(f) Maximum exhaust steam pressure at inlet to exhaust head (see 3.5).
(g) Source of exhaust steam (see 3.5).
(h) Level of preservation and packaging and level of packing required (see 5.1).

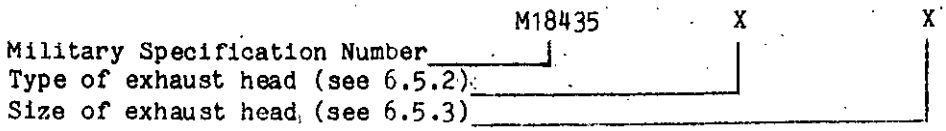
6.3 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractors maintenance and operation manual to be furnished with each exhaust head.

6.4 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-140.55 of the Defense Acquisition Regulation. The first article should be a first production item consisting of one complete exhaust head or it may be a standard production item from the contractor's current inventory a specified in 4.2.1. The contracting officer should include specific instructions in all acquisition instruments, regarding arrangement for examinations, tests, and approval of the first article.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.6 Definitive military specification part number. The military specification part number is a definitive part number which corresponds to the type and size of exhaust heads covered by this specification.

6.6.1 Cataloging data. For cataloging purposes, part numbers for the exhaust heads are assigned as follows:



6.6.2 Type. The type of exhaust head (see 1.2) shall be identified by a single letter:

Type I - A
Type II - B

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6.6.3 Size. The size of the exhaust head (see 1.2) shall be identified by a single letter:

1 - A	8 - L
1-1/4 - B	10 - M
1-1/2 - C	12 - N
2 - D	14 - P
2-1/2 - E	16 - Q
3 - F	18 - R
3-1/2 - G	20 - S
4 - H	24 - T
5 - J	30 - U
6 - K	

Custodian:

Navy - YD

User activity:

Army - CE

Preparing activity:

Navy - YD

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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